RV college of Engineering

Experiential Learning Report

Project-Based Learning

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Name	USN
ABHINAV RACHA BATTUNI	RVCE24BEC178
AMANYA SINHA	RVCE24BEC192
ADITYA SINGH	RVCE24BEC023

Aim:

To design and implement a Smart Irrigation System using a soil moisture sensor to automate watering, save water, and improve plant growth by providing the right amount of water at the right time.

Objectives :

1. To automate the irrigation process using a soil moisture sensor.
2. To save water by ensuring plants are watered only when needed.
3. To reduce manual labor and improve efficiency in watering.
4. To maintain healthy soil moisture levels for better plant growth.
5. To create a cost-effective and user-friendly irrigation system.

Problem Statement :

Traditional irrigation methods often result in overwatering or underwatering, leading to water wastage, poor plant growth, and increased labor costs. There is a need for an efficient, automated system that monitors soil moisture levels and provides the right amount of water to plants, reducing water usage and manual effort.

Experimental Details

Materials Used :

1. Soil Moisture Sensor – Measures the moisture level in the soil.
2. Arduino (e.g., Arduino Uno) – Controls the system and processes data from the sensor.
3. Relay Module – Controls the water pump based on signals from the Arduino.
4. Water Pump – Delivers water to the plants when needed.
5. Jumper Wires – Connects all the components.
6. Breadboard – Used for prototyping and connecting components.
7. Power Supply – Powers the Arduino and the water pump.
8. Water Tubing – Delivers water from the pump to the plants.
9. Resistors – Used for circuit protection.

Methodology:

1. **Setting Up Components:**
   - Connect the soil moisture sensor to the Arduino using jumper wires (VCC, GND, and Signal pins).
   - Attach the relay module to the Arduino to control the water pump.
   - Connect the water pump to the relay module and a water source.
   - Use jumper wires and a breadboard to organize connections.
2. **Programming the Arduino:**
   - Write a code to read soil moisture data from the sensor.
   - Set a threshold moisture level in the code to decide when the pump should turn on/off.
   - Upload the code to the Arduino using a USB cable and the Arduino IDE.
3. **System Testing:**
   - Insert the soil moisture sensor into the soil of a potted plant.
   - Power the Arduino using a power source.
   - Check if the pump activates when the soil is dry and turns off when the soil is wet.
4. **Fine-Tuning:**
   - Adjust the moisture threshold in the code based on plant requirements.
   - Ensure all connections are secure to avoid malfunction.
5. **Implementation:**
   - Set up the system in the field or garden.
   - Test the system over a few days to ensure reliable operation.

Code

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.clear();
  pinMode(2, OUTPUT);
  digitalWrite(2, HIGH);
  delay(1000);
  lcd.setCursor(0, 0);
  lcd.print("IRRIGATION");
  lcd.setCursor(0, 1);
  lcd.print("SYSTEM IS ON ");
    lcd.print("");
    delay(3000);
  lcd.clear();
}

void loop() {
  int value = analogRead(A0);
  Serial.println(value);
  if (value > 950) {
    digitalWrite(2, LOW);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump is ON ");
  } else {
    digitalWrite(2, HIGH);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump is OFF");
  }

  if (value < 950) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : HIGH");
  } else if (value > 300 && value < 950) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : MID ");
  } else if (value > 950) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : LOW ");
  }
}
```

Circuit Diagram

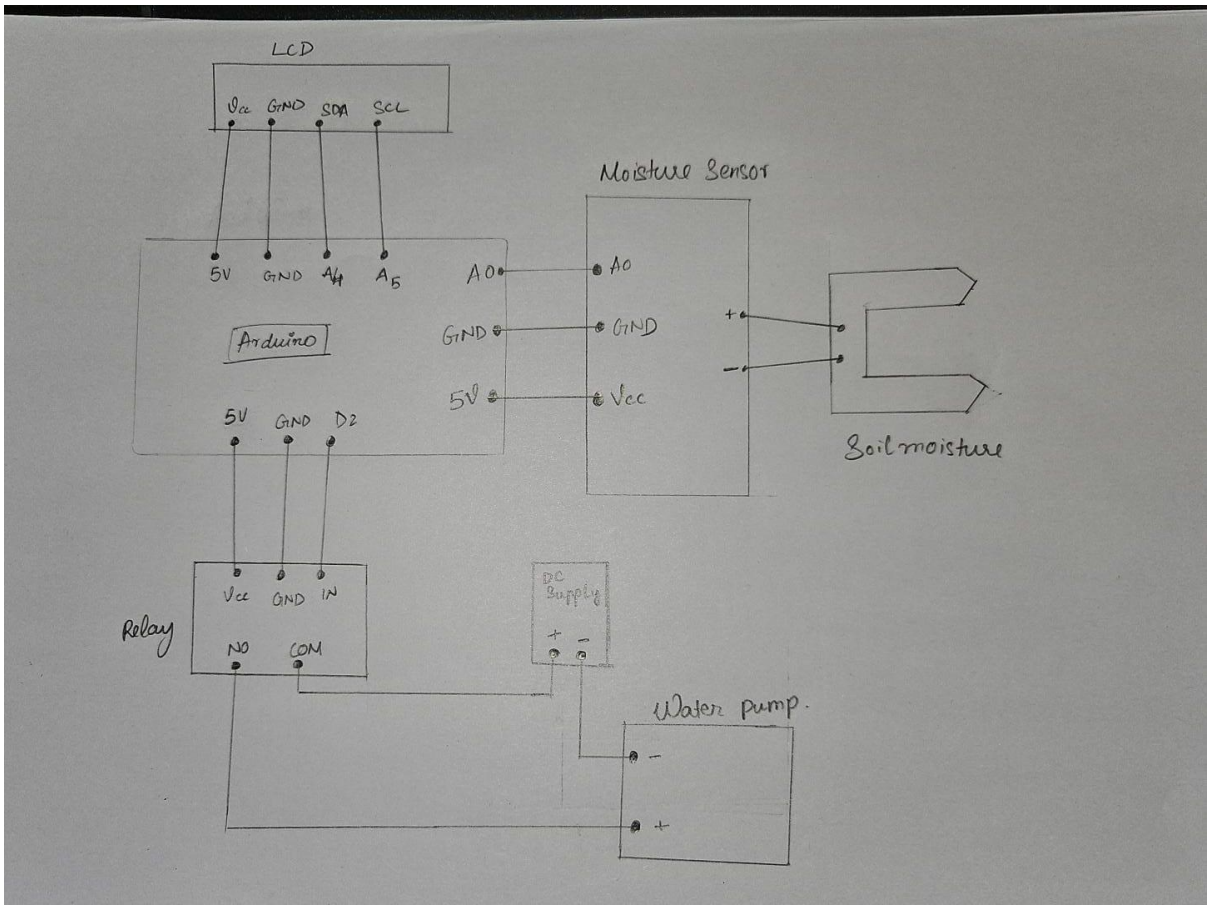

Photo of Model

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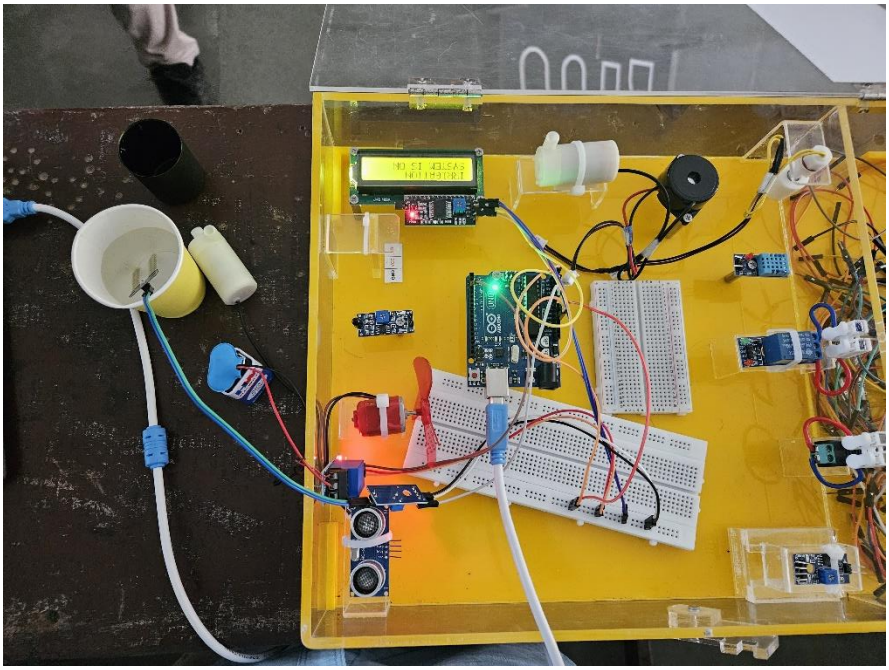

Results and Discussions

Results:

1. **Automated Watering:**
   The system successfully detected soil moisture levels and automatically activated the water pump when the soil was dry.
2. **Water Savings:**
   Compared to manual irrigation, the system reduced water usage by ensuring plants were watered only when necessary.
3. **Improved Plant Health:**
   Plants maintained consistent soil moisture, leading to healthier growth and reduced stress from overwatering or underwatering.
4. **Ease of Use:**
   The system required minimal manual intervention, saving time and labor.
5. **System Reliability:**
   During testing, the system responded accurately to changes in soil moisture levels, making it reliable for long-term use.

Observations:

1. The soil moisture sensor provided accurate readings of soil dryness and wetness.
2. The water pump activated immediately when the soil moisture fell below the threshold.
3. Water was delivered efficiently and stopped once the required moisture level was achieved.
4. The system operated without manual intervention, reducing human effort.
5. Occasional calibration of the sensor was needed to ensure accuracy, especially in different soil types.
6. Power supply interruptions affected the system's functionality temporarily.
7. The overall system worked better in smaller setups like potted plants or small gardens during initial testing.

Inferences

1. The Smart Irrigation System is effective in automating the watering process, ensuring optimal soil moisture levels.
2. The system reduces water wastage by only activating the water pump when necessary.
3. Automation minimizes manual labor, making it suitable for busy farmers or gardeners.
4. Proper calibration of the soil moisture sensor is crucial for consistent performance across different soil types.
5. The system is reliable for small-scale applications and can be scaled up with additional sensors and pumps for larger fields.
6. A stable power supply is essential for uninterrupted operation.
7. The project demonstrates a practical and affordable solution for sustainable water management in agriculture.

Conclusion

The Smart Irrigation System successfully automated the watering process by using a soil moisture sensor to monitor and maintain optimal soil moisture levels. This reduced water wastage and ensured healthy plant growth while minimizing manual effort. The system proved to be reliable and cost-effective, making it suitable for small-scale agricultural or gardening applications. Challenges such as sensor calibration and power interruptions were observed but can be addressed with further improvements. Overall, the project demonstrated a sustainable approach to efficient water management in agriculture.